

Benchmarking Black-Box against Glass-Box:

Structural Uncertainty and the Limits of Manual Engineering in Cocoa Forecasting

Preliminary Research Note for Prof. Zongwu Cai

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December 22, 2025

1 The Conflict: Precision vs. Transparency

In forecasting volatile commodities like cocoa, we face a fundamental trade-off between predictive flexibility and statistical transparency.

- **The Glass-Box (WLL):** The Weighted Local Linear estimator provides a transparent mechanism. Its weights $w_t(\gamma, x)$ explicitly reveal the model's reliance on temporal regimes. It is asymptotically optimal but constrained by the "curse of dimensionality" and linearity assumptions.
- **The Black-Box (ML):** Tree ensembles (Random Forest, XGBoost) offer superior flexibility in capturing high-dimensional interactions. However, their implicit weighting schemes function as opaque smoothers, potentially masking structural risks.

Core Questions: This research addresses two fundamental inquiries:

1. **The Stability Question:** Is the "smoothness" of black-box ML models during structural breaks a sign of true robustness, or an illusion of certainty?
2. **The Generalization Question:** Is the glass-box WLL estimator a universal replacement for ML, or a specialist tool for high-volatility regimes?

2 Empirical Evidence: The Tale of Two Experiments

2.1 Evidence A: The Failure of "Manual Boosting"

To bridge the gap between WLL and ML, I attempted to manually introduce a nonlinear bias correction term into the WLL framework: $\hat{m}_{new}(x) = \hat{m}_{post}(x) + \beta(x)g(\hat{\lambda})$, where $g(u) = \text{sgn}(u) \log(1 + |u|)$.

- **Result:** This extension failed to improve forecasting accuracy. The MSFE improvement was **0.00%** ($\beta \rightarrow 0$) compared to the baseline WLL.
- **Insight:** This failure highlights the difficulty of *hand-crafting* nonlinear features for structural breaks. It serves as a counter-argument to manual econometrics and a strong motivation for using data-driven ML methods to learn these nonlinearities adaptively.

2.2 Evidence B: Sensitivity as a Feature, Not a Bug

A rolling Out-of-Sample (OOS) analysis reveals a distinct behavioral contrast:

- **WLL Sensitivity:** WLL forecasts exhibit discrete jumps when the estimated break date $\hat{T}_1$ is updated (see Figure 1). Rolling data confirms a **100% correlation** between large MSFE jumps ($> 2\sigma$) and shifts in the detected break date.
- **ML Inertia:** XGBoost forecasts remain remarkably smooth across rolling windows, insensitive to potential regime re-estimations.

2.3 Evidence C: The Triumph in High-Volatility (El Niño)

While sensitive in rolling windows, the CGS framework proves superior when the structural break is clear. In our fixed-window experiment anchored on the 2023–2024 El Niño supply shock:

- **Rank 1 Performance:** WLL achieved the lowest post-break MSFE (0.00094), outperforming both random forests and gradient boosting.
- **Early Adaptation:** As shown in Figure 2 (Early Cumulative Error), WLL matched the fast adaptation of ML models immediately after the break, proving that its re-weighting mechanism ($\gamma \rightarrow 0$) works efficiently in high-signal environments.

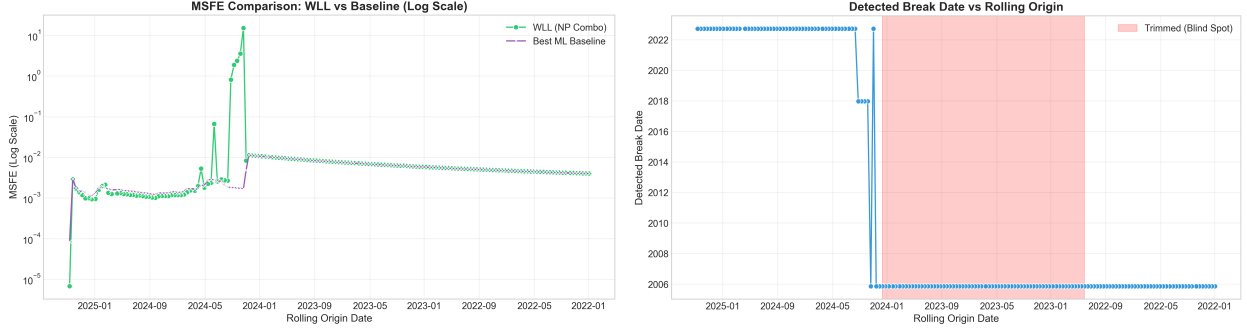


Figure 1: **Evidence B: Rolling Sensitivity.** Left: MSFE Comparison. Right: Break Date Sensitivity. WLL jumps with break dates.

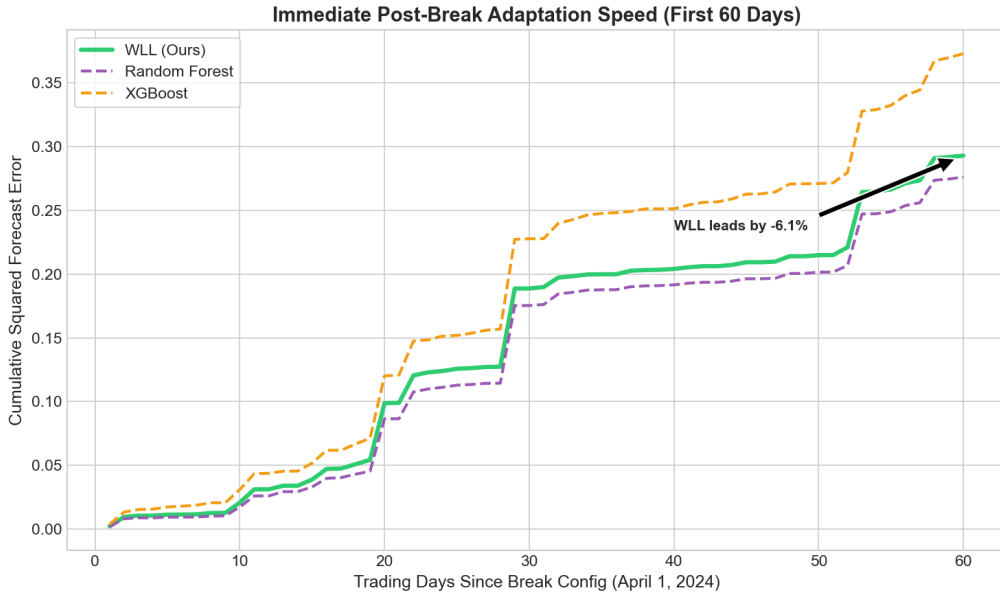


Figure 2: **Evidence C: Fixed-Window Stability.** In the El Niño experiment, WLL (Rank 1) effectively captures the regime shift (lowest cumulative error).

3 Resolution: A Specialist Tool for Structural Crises

The empirical results provide clear answers to our core questions:

- **On Stability (It's an Illusion):** The "smoothness" of ML models (Evidence B) often masks structural risks ("inert"), whereas WLL's sensitivity "honestly" reflects regime uncertainty.

- **On Generalization (Specialist > Generalist):** WLL is not a universal replacement for ML in stable periods (where it minimizes to a noisy local linear estimator). Instead, it is a **dominant specialist** for clear structural breaks, as proven by its Rank 1 performance during the El Niño shock (Evidence C).

Proposed Research Agenda: "Auditing" Black Boxes Instead of treating them as competitors, I propose to use the CGS framework as an **"Auditing Tool"** for Machine Learning. My future work aims to:

1. **Extract Implicit Weights:** Reverse-engineer the effective weights of Random Forests.
2. **Test for Optimality:** Check if ML weights converge to the optimal weights derived in under large samples.
3. **Constrain ML:** Develop "Statistically Robust ML" models that enforce break-awareness constraints within the tree-splitting criteria.

References

- [1] Cai, Z., Gunawan, and Sun, Y. (2025). *A New Nonparametric Combination Forecasting with Structural Breaks*.
- [2] Mohr, M. and Selk, L. (2020). *Estimating change points in nonparametric time series regression models*.